Analysis on World Happiness Dataset

Presented by Zaw Min Tun

Given dataset

World Happiness Dataset:

	Country	Happiness_Score	GDP_per_Capita	Social_Support	Healthy_Life_Expectancy	Freedom_to_Make_Choices	Generosity	Perceptions_of_Corruption
1	Norway	6.25	1.39	0.82	81.6	0.69	0.01	0.71
2	Denmark	3.61	1.27	0.43	66.9	0.48	0.43	0.53
3	Iceland	4.68	0.87	0.54	63.4	0.71	0.41	0.72
4	Switzerland	4.46	0.67	0.57	68.8	0.93	0.32	0.52
5	Finland	6.67	1.55	0.45	75.4	0.58	0.16	0.10
6	Netherlands	6.41	0.87	0.54	72.6	0.45	0.38	0.36
7	Canada	7.34	0.60	0.46	49.6	1.00	0.07	0.12
8	New Zealand	3.87	0.61	0.57	41.3	0.66	0.26	0.84
9	Australia	5.31	1.43	0.78	53.2	0.36	0.27	0.80
10	Sweden	3.63	1.16	0.62	51.2	0.33	0.57	0.77
11	United States	4.44	1.39	0.62	48.9	0.38	0.53	0.35
12	United Kingdom	5.67	1.30	0.53	79.6	0.74	0.16	0.15
13	Germany	3.61	1.09	0.56	76.8	0.85	0.30	0.80
14	France	4.36	1.57	0.96	53.2	0.60	0.11	0.86
15	Japan	6.29	0.92	0.79	67.5	0.34	0.55	0.17
16	South Korea	5.84	1.11	0.77	56.6	0.57	0.52	0.49
17	India	4.45	1.41	0.50	78.4	1.00	0.18	0.16
18	China	6.03	1.18	0.84	59.3	0.67	0.38	0.71
19	Brazil	6.98	1.45	0.50	51.1	0.98	0.37	0.71
20	South Africa	3.53	1.14	0.63	50.4	0.90	0.09	0.20

Exploration & check:

Missing values & duplicates

```
--- Missing Values (per column) ---
Country                        0
Happiness_Score                0
GDP_per_Capita                 0
Social_Support                 0
Healthy_Life_Expectancy        0
Freedom_to_Make_Choices        0
Generosity                     0
Perceptions_of_Corruption      0
dtype: int64
No missing values found.
```

```
--- Duplicate Rows ---
Number of duplicate rows: 0
No duplicates found.
```

Exploration & check:

Data types

```
--- Data Types ---
Country                        object
Happiness_Score                float64
GDP_per_Capita                 float64
Social_Support                 float64
Healthy_Life_Expectancy        float64
Freedom_to_Make_Choices        float64
Generosity                     float64
Perceptions_of_Corruption      float64
dtype: object
```

Exploration & check:

Basic statistics

```
--- Basic Statistics ---
       Happiness_Score  GDP_per_Capita  Social_Support  Healthy_Life_Expectancy  Freedom_to_Make_Choices  Generosity  Perceptions_of_Corruption
count        20.000000       20.000000       20.000000                20.000000                20.000000   20.000000                  20.000000
mean          5.171500        1.149000        0.624000                62.290000                 0.661000    0.303500                   0.503500
std           1.250715        0.304785        0.150382                12.332708                 0.227617    0.171196                   0.277532
min           3.530000        0.600000        0.430000                41.300000                 0.330000    0.010000                   0.100000
25%           4.237500        0.907500        0.522500                51.175000                 0.472500    0.160000                   0.192500
50%           4.995000        1.170000        0.570000                61.350000                 0.665000    0.310000                   0.525000
75%           6.260000        1.395000        0.772500                73.300000                 0.862500    0.415000                   0.732500
max           7.340000        1.570000        0.960000                81.600000                 1.000000    0.570000                   0.860000
```

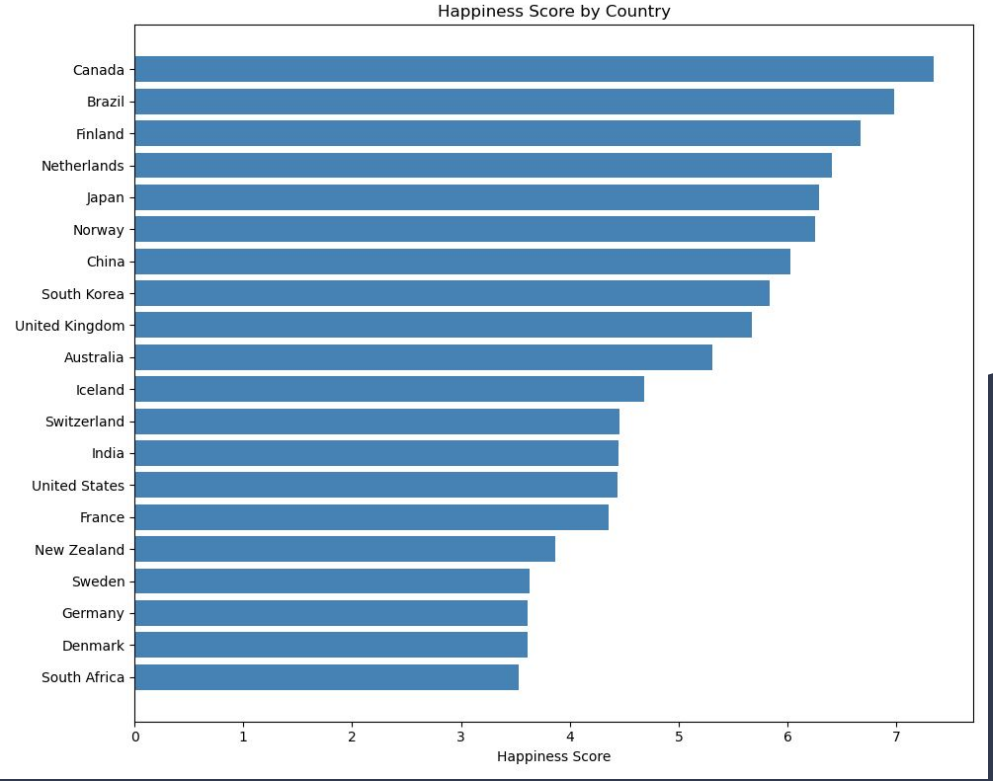
Exploration & check:

Outliers

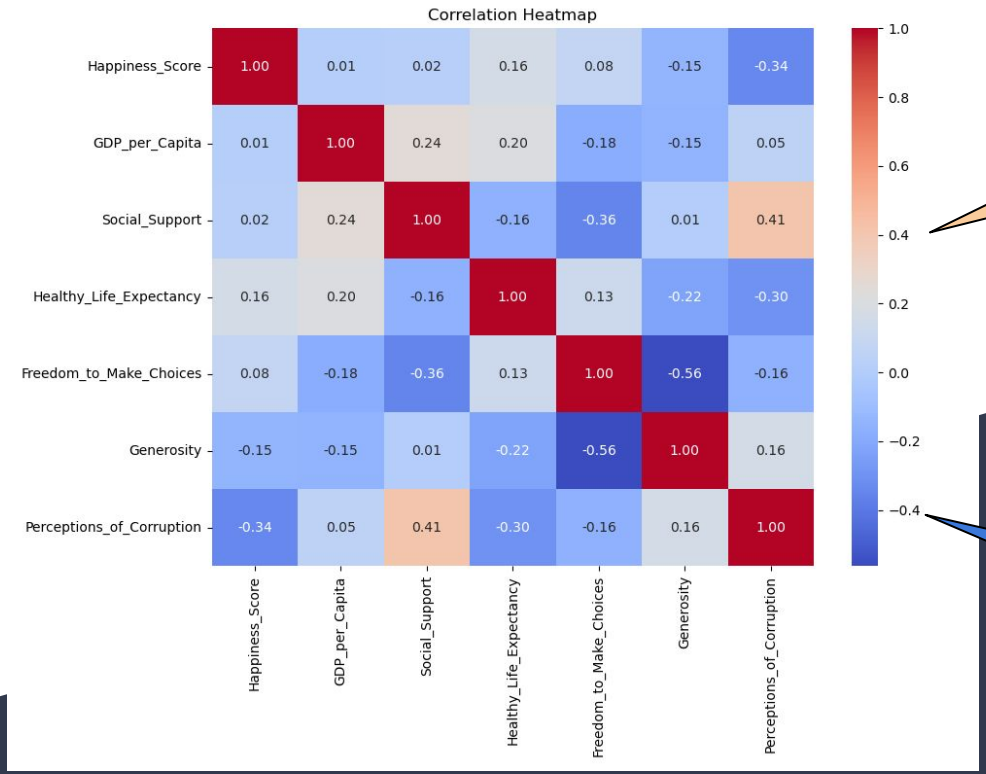
```
--- Outliers (IQR method, per column) ---
Happiness_Score                0
GDP_per_Capita                 0
Social_Support                 0
Healthy_Life_Expectancy        0
Freedom_to_Make_Choices        0
Generosity                     0
Perceptions_of_Corruption      0
dtype: int64
No outliers detected.
```

IQR is calculated as 75% - 25%.
For example, for Happiness_Score, it is 2.02 (6.26 - 4.24).
Any value below (Q1 - (1.5 × IQR)), or above (Q3 + (1.5 × IQR)) is flagged as an outlier.

Visualization: Horizontal Bar Chart



Visualization: Heatmap



‘Social support’
vs
‘Perceptions of corruption’

‘Generosity’
vs
‘Freedom to make choices’

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Thank you!!!